# Jhanvi Cars — Project Dependency Guide

Frontend, backend, database, authentication, development, containerization, and DevOps dependencies.

## 1. Project Architecture

**Frontend:** React + TypeScript + Vite
**Backend:** Node.js + TypeScript + Express microservices
**Database:** PostgreSQL
**Database access:** Knex + pg
**Containerization:** Docker
**Orchestration:** Kubernetes
**CI/CD:** GitHub + Jenkins + Docker Registry
**Observability:** Prometheus + Grafana + Loki + Alloy
**Infrastructure:** Terraform + AWS

## 2. Backend Microservices

| Service | Role |
|---|---|
| auth-service | Authentication, registration, login, JWT and password handling |
| user-service | User-related functionality |
| driver-service | Driver profiles, availability and vehicle information |
| vehicle-service | Vehicle-related functionality |
| booking-service | Ride/booking lifecycle |
| payment-service | Payment-related functionality |
| notification-service | Notifications |

## 3. Confirmed Auth-Service Runtime Dependencies

| Dependency | Purpose |
|---|---|
| Node.js 20 | JavaScript/TypeScript runtime |
| Express | HTTP server and REST API framework |
| dotenv | Loads environment variables |
| Zod | Request/input validation |
| bcrypt | Password hashing |
| jsonwebtoken | JWT authentication tokens |
| Knex | SQL query builder and database migrations |
| pg | PostgreSQL driver |
| uuid | Unique ID generation |
| pino | Structured application logging |
| pino-http | HTTP request logging |

## 4. Backend Development Dependencies

| Dependency | Purpose |
|---|---|
| TypeScript | Compiles TypeScript into JavaScript |
| ts-node | Runs TypeScript directly |

| Dependency | Purpose |
|---|---|
| ts-node-dev | Development server with automatic restart |
| Jest | Testing framework |
| Supertest | HTTP/API testing |
| ts-jest | TypeScript support for Jest |
| @types/* | TypeScript type definitions |

## 5. Database Stack

**PostgreSQL** is the relational database. **pg** provides the Node.js PostgreSQL driver and **Knex** provides query building and migrations.

```
Node.js → Knex → pg → PostgreSQL
```

The driver service has migrations including the vehicle-detail migration adding vehicle_type and vehicle_model. The migration status was verified with no pending migrations.

## 6. Authentication Stack

**bcrypt** handles password hashing and **jsonwebtoken** handles JWT-based authentication. The frontend sends credentials to the auth service, receives a token, and uses it for authenticated API requests.

```
Frontend → Auth Service → credential verification → JWT → authenticated API requests
```

## 7. Validation

**Zod** validates incoming service requests. Driver registration validates vehicle type and vehicle model. Supported vehicle types are AUTO, ECONOMY, COMFORT and COMFORT_PLUS.

## 8. Logging and Testing

**Pino** and **pino-http** provide structured application and HTTP logging. **Jest**, **Supertest** and **ts-jest** provide the confirmed auth-service testing stack.

## 9. Docker

The confirmed auth-service Dockerfile uses a multi-stage Node 20 Alpine build. The build stage installs dependencies and runs the TypeScript build. The runtime stage installs production dependencies, copies the compiled application, runs as a non-root user, and starts dist/server.js.

```
Build stage → npm ci → npm run build → Runtime stage → production dependencies → non-root container
```

## 10. DevOps / Infrastructure Stack

| Technology | Role |
|---|---|
| GitHub | Source control and collaboration |
| Jenkins | CI/CD automation |
| Docker | Containerization |
| Docker Registry / ECR | Container image storage |
| Kubernetes | Container orchestration |
| Terraform | Infrastructure as Code |
| AWS | Cloud infrastructure |
| Prometheus | Metrics collection |
| Grafana | Metrics visualization |

| Technology | Role |
|---|---|
| Loki | Log aggregation |
| Alloy | Telemetry/log collection |
| Nginx / Ingress | HTTP routing and reverse proxying |
| cert-manager / Let's Encrypt | TLS certificate automation |
| Calico | Kubernetes networking |

## 11. Dependency Map

Frontend → API layer → Microservices → PostgreSQL
Microservices → Express + TypeScript + Zod + Knex + pg
Authentication → bcrypt + JWT
Logging → Pino + pino-http
Testing → Jest + Supertest + ts-jest
Packaging → Docker
Orchestration → Kubernetes
CI/CD → GitHub + Jenkins + Docker Registry/ECR
Observability → Prometheus + Grafana + Loki + Alloy
Infrastructure → Terraform + AWS

## 12. Scope Note

This guide includes dependencies and architecture confirmed during the current project review. The auth-service package.json was inspected in detail; the seven backend service names were confirmed, but the other six service package.json files were not yet inspected, so their service-specific npm dependencies are intentionally not presented as confirmed facts.